Interdisciplinary Project in Data Science (2025)

Constructing a Reddit-based Inflation Index

Sentiment-driven text analysis of price-related discussions across subreddits

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Abstract

This project investigates whether Reddit can serve as a useful source for creating a sentiment-based inflation index. Using historical Reddit data (2012–2024) from selected economics- and consumer-oriented subreddits, I apply three modeling approaches to extract sentiment signals: a rule-based method (VADER), a weakly supervised Transformer model trained on Reddit post scores, and a fully supervised regression model that directly predicts consumer price index values. The evaluation is based on official US inflation data at the quarterly, semi-annual, and annual aggregation levels. The results show that the selection of subreddits and temporal aggregation are crucial for performance:`povertyfinance` and `food` achieved the strongest correlation with the CPI value. Among the models, the weakly supervised BERT approach outperforms VADER and the direct CPI regression, achieving correlations of over r = 0.95 with minimal error rates. These results demonstrate the feasibility of a Reddit- based inflation sentiment index and underscore the potential of social media as a proxy for traditional inflation measurements. To make these findings more accessible and implement them in practice, an interactive dashboard was built that calculates the index on a daily, weekly, or monthly basis using up-to-date Reddit data. It uses the most powerful model (BERT with score supervision) and the most reliable subreddits (`povertyfinance`, `food`) identified in the experiments.

Contents

1 Introduction

Understanding inflation dynamics in near real time is crucial for effective monetary policy and economic decision-making. Official indicators such as the consumer price index (CPI) are published with a delay, so alternative data sources can be used to supplement traditional methods. Various studies have demonstrated the potential of social media in this context. For example, [7] created a high-frequency inflation sentiment index based on Twitter data and demonstrated its predictive power compared to official measures. [2] uses Reddit data to forecast inflation and unemployment in the eurozone. Similar research shows the role of text-based data in predicting macroeconomic indicators [11, 1].

Parallel studies have shown that internet behavior, in particular, search volume on Google—contains predictable information about inflation expectations. [4] conducted a real-time measurement of inflation expectations based on Google search data and demonstrated its potential for predicting inflation. Similar results have also been obtained in other countries, including India [6] and Russia [8], where online search queries and user content on social media were strongly correlated with inflation. Despite these findings, Reddit remains relatively unexplored in this context. Reddit is a social media platform that organizes discussions into topic-specific forums called “subreddits.” Posts and comments can be upvoted or downvoted by users, providing a simple, community-based relevance signal. Subreddits such as `personalfinance` or `povertyfinance` contain potentially valuable discussions about everyday (economic) experiences that are likely to reflect real-time concerns about inflation. Therefore, this project investigates whether sentiment indices can be extracted from Reddit data that can serve as a valid, timely indicator of inflation.

2 Data Collection

2.1 Data Source

At the outset, three data sources were considered, with the third source being selected: (1) Pushshift API: Although this API provides convenient access to Reddit data, it has not been reliable since mid-2023 due to server issues and a lack of updates. (2) Official Reddit API: The official API does not allow access to historical data older than a few months, making it unsuitable for creating long-term inflation indices. (3) Academic Torrents: To access the complete historical data, I used **Academic Torrents**[1]. This dataset covers the period **from June 2005 to December 2024** and is more than **3.1 TB** in size.

Each file contains one month’s worth of data in `.zst`-compressed, line-separated JSON format (NDJSON). Each line corresponds to a single Reddit post or comment object. For scalable processing, a custom Python extraction script was implemented that filters Reddit posts based on several criteria to ensure the relevance and quality of the content. Only posts from specific target subreddits and within a clearly defined time window are considered. In addition, entries are excluded if their score falls below a predefined threshold (here 1) or if they do not contain any comments in order to filter out content with low relevance. To ensure thematic relevance, the title and text are checked for the presence of economic or inflation-related terms using a regular expression. This curated keyword set was defined in the configuration file as shown below:

```
{
  "limit": 10000,
  "min_score": 1,
  "min_comments": 0,
  "keywords":
    ["prices", "expensive", "too expensive", "went up", "cost", "costs", "more expensive",
    "inflation", "price increase", "groceries", "grocery", "shopping", "bills", "budget",
    "rent", "housing", "landlord", "utilities", "electricity", "gas bill", "heating",
    "water bill", "food", "eggs", "milk", "bread", "meat", "vegetables", "fruit", "toilet
    paper", "gas", "gasoline", "fuel", "afford", "broke", "struggling", "unaffordable",
    "cheaper", "saving money", "cutting back", "walmart", "povertyfinance", "personalfinance",
    "frugal", "budgeting", "cost of living", "price hike", "price gouging", "consumer prices",
```

[1] https://academictorrents.com/details/ba051999301b109eab37d16f027b3f49ade2de13

```
    "inflation rate", "economic hardship"]
}
```

The keywords were selected heuritretically. Min score equal to 1 ensures a good trade-off between a large amount of data and qualitatively relevant poreas. min_comments was left at 0 because this would have limited the amount of data too much.

2.2 Subreddits

The selection of subreddits was also carried out heuristically with systematic intent. The goal was to ensure coverage across different community sizes. All chosen subreddits exhibit a thematic connection to either product prices, economic concerns, or personal budgeting, thus serving as suitable candidates for capturing inflation-related sentiment.

- **r/food** (24.4M): Discussions around food prices in restaurants, grocery stores, delivery services.
- **r/personalfinance** (21.2M members): Subreddit with discussions about prices, living costs.
- **r/frugal** (6.6M): Users share money saving strategies and complain about rising prices.
- **r/economy** (1M):Valuable for contextualizing price developments from a policy or news angle.
- **r/povertyfinance** (2.3M): Posts describe direct economic struggles related to rent, food, utility bills.
- **r/Costco** (1.3M): Focused on product prices and consumer experiences within a U.S. retail chain.
- **r/AskAnAmerican** (1.1M): Provides insights into U.S. life, including questions about affordability.
- **r/walmart** (350k): Covers pricing trends, product availability within another major retail setting.

In order to keep the data volume manageable during the initial development phase, only data for the month of December from 2016 to 2024 was extracted. This development set allows for a comparison of the eight subreddits, so that only the most relevant subreddits for further modeling are filtered out at an early stage.

2.3 Timeperiod and Inflationsindex

The full training dataset covers the period from March 2012 to December 2024, with data collected quarterly, i.e., in March, June, September, and December of each year. Since the original dataset comprises over 3.1 TB of compressed Reddit content, using all available months would not be computationally feasible and is not necessary for the scope of this project. Quarterly sampling offers a robust yet practical compromise that allows the central hypothesis to be tested efficiently. For inflation values, the project uses the official CPI from the U.S. Bureau of Labor Statistics (BLS). The monthly CPI data is available at `https://fred.stlouisfed.org/series/CPIAUCNS`. Figure 1 shows the data volume. For example, you can see that `r/personalfinance` only starts in 2016. Other subreddits that start in 2016 only start in that year because they were pre-filtered by an initial performance test (Model 1) and no further data was downloaded.

3 Models

This section presents three selected modeling approaches:

- **Model 1: VADER** — A fast, rule-based sentiment model using a fixed polarity lexicon, widely used for social media sentiment analysis [5].
- **Model 2: BERT (Score Supervision)** — A weakly-supervised transformer model trained to distinguish highly upvoted from downvoted economic posts. Transformer-based models capture contextual nuances better than dictionary methods [3, 10], and have been successfully applied to inflation-related sentiment analysis on Twitter and other platforms [7].
- **Model 3: BERT (Inflation Supervision)** — A fully supervised regression model that directly predicts monthly inflation rates based on aggregated post content. This approach is in line with recent

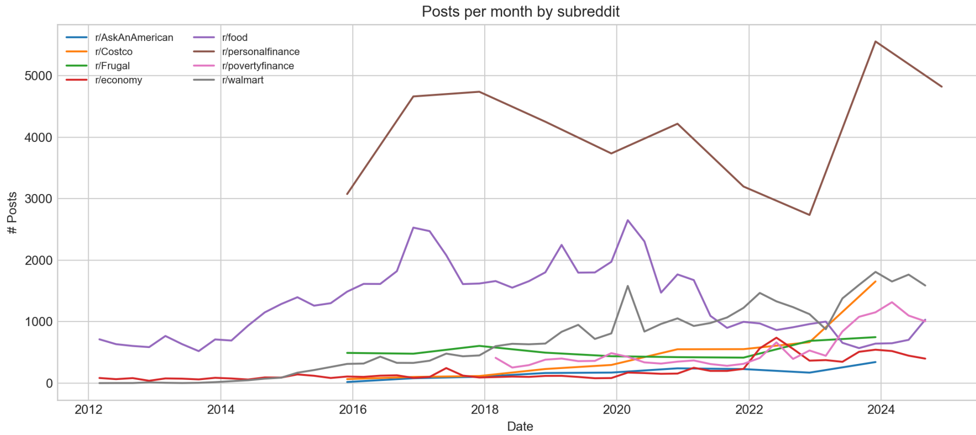

Figure 1: Number of Reddit posts across selected subreddits

advances using transformer-based and LLM-driven sentiment analysis for nowcasting and forecasting macroeconomic variables such as inflation [2, 11].

3.1 Model 1: VADER

To calculate an initial sentiment index from the Reddit data, I used VADER (Valence Aware Dictionary and sEntiment Reasoner). VADER is a lexicon- and rule-based sentiment analysis model that was developed specifically to capture the polarity of short texts in social media. It returns four ratings: `positive`, `neutral`, `negative`, and a composite `compound` rating between -1 and $+1$, which summarizes the overall sentiment of the text. The compound rating was used for this project. It is calculated as the normalized weighted sum of the valence values of each word in the text. The final sentiment value is calculated as follows:

$$\text{compound} = \frac{x}{\sqrt{x^2 + \alpha}}, \quad \text{where } x = \sum \text{valence terms}, \ \alpha = 15$$

Consider the sentence: *"I'm extremely happy with the lower food prices!"*

VADER assigns the following valence scores to the relevant words (Demonstrative values):

- `happy` $\rightarrow +3.2$
- `extremely` $\rightarrow$ intensifier, boosts valence by 0.293

Thus, the adjusted valence score x becomes $3.2 \times (1 + 0.293) = 4.14$.

Plugging into the formula:

$$\text{compound} = \frac{4.14}{\sqrt{(4.14)^2 + 15}} = \approx 0.73$$

This results in a positive sentiment score of `0.73`. In addition, I developed a script to process all filtered Reddit posts per subreddit and month. For each post, I merged the fields `title` and `selftext` and calculated the score using VADER. The sentiment scores were then aggregated per month by calculating the arithmetic mean of all posts:

$$\text{avg\_sentiment}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \text{compound}_i$$

where N_t is the number of posts in month t of a specific subreddit. [5]

3.2 Model 2: BERT (Score Supervision)

To complement the rule-based VADER approach, I implemented a two-stage pipeline based on a pre-trained Transformer model (`distilbert-base-uncased`). Instead of using external labels, this model uses Reddit's built-in voting mechanism as a proxy for collective sentiment. Unlike the first model, Model 2 with BERT can better capture context, which can be important because posts on the topic of inflation may contain sarcasm and irony that are misinterpreted by simple keyword lists. The assumption is that Reddit users express their opinions through upvotes and downvotes. Posts that describe economic problems (e.g., high prices, rent increases, etc.) and receive many upvotes likely reflect agreement about rising costs. Conversely, posts that do not receive upvotes may express sentiments that are either less relevant. Contributions with a score of ≥ 3 are marked as *positive*, while those with ≤ 2 are marked as *negative*. The model thus learns linguistic patterns that are typical of supported or rejected economic narratives. The three were chosen heuristically and can be varied in further performance optimization measures.

Example 1:

`"My grocery bill has doubled in the last two months, I can't keep up."`
$\rightarrow$ Score: 42 $\Rightarrow$ Labeled as **positive** (high resonance)
The model learns that strong emotional language about rising costs, paired with personal struggle, is typical for high-scoring posts.

Example 2:

`"Rent is fine if you just budget better."`
$\rightarrow$ Score: 1 $\Rightarrow$ Labeled as **negative** (low resonance)
The model learns that dismissive language is negatively evaluated and thus differs semantically.

After training, the model is applied to all posts in a given subreddit. It predicts for each post whether it is linguistically closer to the "positive" or "negative" class, as learned in Stage 1. These predictions are then averaged per month to form a sentiment curve. Formally, for each month m, the sentiment index is computed as:

$$\text{SentimentIndex}_m = \frac{1}{N_m} \sum_{i=1}^{N_m} \hat{y}_i$$

where N_m is the number of classified posts in month m, and $\hat{y}_i \in \{0, 1\}$ is the predicted label for post i. This yields a index score between 0 and 1, where a rising curve implies that more posts match the linguistic style of high-scoring economic complaints, potentially indicating growing inflation-related concern. declining curve suggests fewer such resonant expressions, possibly reflecting a less critical economic sentiment. The transformer-based approach builds on the original Transformer architecture [12], the BERT framework [3], its distilled variant DistilBERT [10], and the principles of weak supervision [9]. Due to the high computational costs, this pipeline was limited to the four best subreddits based on the performance of the VADER model. Among these four, the two subreddits with the best performance were used for the third model.

3.3 Model 3: BERT (CPI Regression)

Compared to the two previous models, which derive an index value and then compare it with the official inflation data, model 3 learns directly from the inflation values. For each month, all Reddit posts from a specific subreddit are summarized in a single text document. This document serves as input X_m, while the CPI for the same month serves as the target value y_m. The task is to learn a function $f : X \rightarrow y$ that estimates inflation directly from the text. Formally, the model learn a f_θ such that:

$$\hat{y}_m = f_\theta(X_m)$$

where X_m is the aggregated Reddit text from month m, and $\hat{y}_m$ is the model's predicted CPI value. Before training, CPI values are standardized:

$$y_m^{(norm)} = \frac{y_m - \mu}{\sigma}$$

with μ and σ as the mean and standard deviation of the CPI values. I used a transformer-based language model (`distilbert-base-uncased`) that was developed to minimize the mean square error (MSE) between the predicted and index values. The model learns which semantics are associated with high or low inflation. For example, months with expressions such as:

```
"My rent increased by 30% this year and groceries are insane."
```

consistently associated with higher CPI values, the model learns to associate such language patterns with high inflation. Conversely, if posts contain more stable or optimistic tones:

```
"Finally seeing gas prices go down, things might be normalizing."
```

This approach builds on the Transformer architecture [12], the BERT framework [3] and its distilled variant DistilBERT [10].

4 Results and Evaluation

4.1 VADER Model

4.1.1 Development-Set Evaluation on Quarterly Data

In order to determine an initial correlation between the Reddit-based sentiment indicators and the official US inflation data with as little computational effort as possible, I used a development data set. For each subreddit, I calculated the sentiment indices for the twelfth month in the years 2016 to 2024 and compared them with each other. In addition, a combined index was created in which the sentiment values of all subreddits are aggregated based on a monthly average weighted by the number of posts.

Figure 2 compares the standardized monthly sentiment indices from the VADER model (blue) with the standardized CPI values for December (orange) for each of the eight selected subreddits and the aggregated combined index. The correlation coefficients (r), mean absolute errors (MAE), and root mean square errors (RMSE) vary considerably between subreddits, showing that the selection of subreddits is an important factor in the agreement between sentiment indicators and official inflation figures.

Several subreddits show a high correlation, suggesting that their discourse patterns capture relevant price signals, in particular `economy` ($r = 0.88$, MAE $= 0.41$, RMSE $= 0.46$), `food` ($r = 0.78$, MAE $= 0.53$, RMSE $= 0.62$), `povertyfinance` ($r = 0.87$, MAE $= 0.61$, RMSE $= 0.46$), and `walmart` ($r = 0.63$, MAE $= 0.74$, RMSE $= 0.81$). Due to their strong performance, these four subreddits were selected for the subsequent models. Limiting further model development to these sources enables a significant reduction in memory and computing requirements while retaining the most meaningful sentiment signals. In later phases, these subreddits will be expanded on a quarterly basis, starting in 2012, in order to capture finer dynamics.

Other subreddits perform significantly worse, for example `costco` ($r = 0.12$), `AskAnAmerican` ($r = 0.45$) and `personalfinance` ($r = 0.33$). To assess whether the size of the subreddits could explain the differences in the correlation and error metrics, Pearson correlations between the number of subreddit members and the values for r and MAE were calculated. This analysis helps determine whether larger communities inherently generate more representative or less noisy sentiment signals, or whether the observed differences are primarily topic-specific rather than size-dependent. The results are visualized in Figure 3. The left panel shows the relationship between the size of the subreddit and the MAE, with the Pearson correlation coefficient close to zero ($r = 0.02$, $p = 0.968$), suggesting that there is no significant correlation between the number of members and the deviation between sentiment and CPI. The right-hand field examines the relationship between the size of the subreddit and the correlation coefficient r. Here, too, the correlation is negligible ($r = -0.01$,

Figure 2: Z-normalized sentiment index (VADER) vs. CPI for each subreddit (December months, 2016–2024).

$p = 0.976$), which means that the size of the community has no systematic influence on the linear relationship between sentiment indices and CPI trends.

4.1.2 Quarterly Data (Full-Set)

To assess robustness, I examine the correlation between sentiment and the official consumer price index (CPI) across three temporal resolutions:

1. **Quarterly**, where only the final month of each quarter (March, June, September, December) is used as a representative snapshot.
2. **Semi-annual**, calculated as the average of two quarterly values (Q1+Q2 and Q3+Q4).
3. **Annual**, defined as the average of both semi-annual values over a year.

For simplicity, the term *quarterly* will be used throughout the remainder of this report to refer to this specific definition (i.e., end-of-quarter snapshots), unless otherwise stated. This stepwise aggregation makes it possible to assess how the predictive power of the correlation changes as the temporal granularity is reduced.

Figure 4 shows the quarterly results for the four best-performing subreddits from the development set (`economy`, `food`, `povertyfinance`, `walmart`) applied to the **full dataset**. Overall, the correlations and error metrics are largely consistent with the results of the development set.

- **economy**: $r = 0.20$, MAE $= 0.96$, RMSE $= 1.26$. Correlation drops compared to the development set, suggesting increased noise in quarterly resolution, possibly due to higher intra-year variability.
- **food**: $r = 0.64$, MAE $= 0.61$, RMSE $= 0.84$. Maintains strong positive correlation with moderate errors, consistent with dev set.

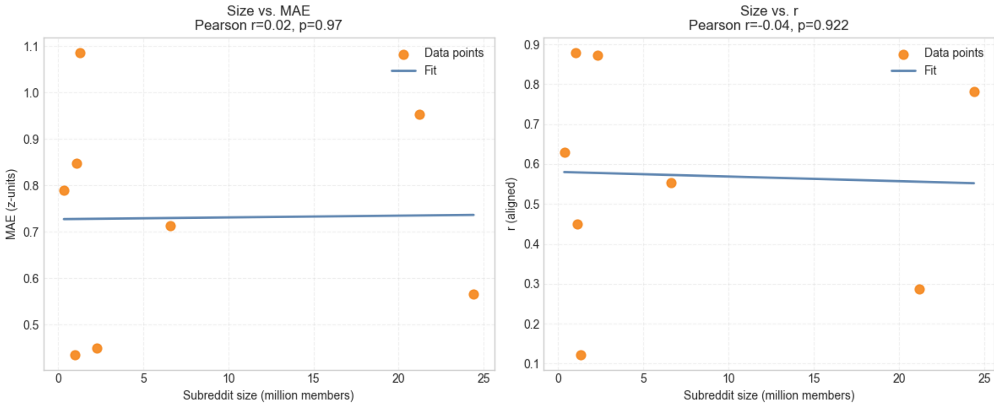

Figure 3: Relationship between subreddit size and (left) MAE and (right) correlation coefficient r

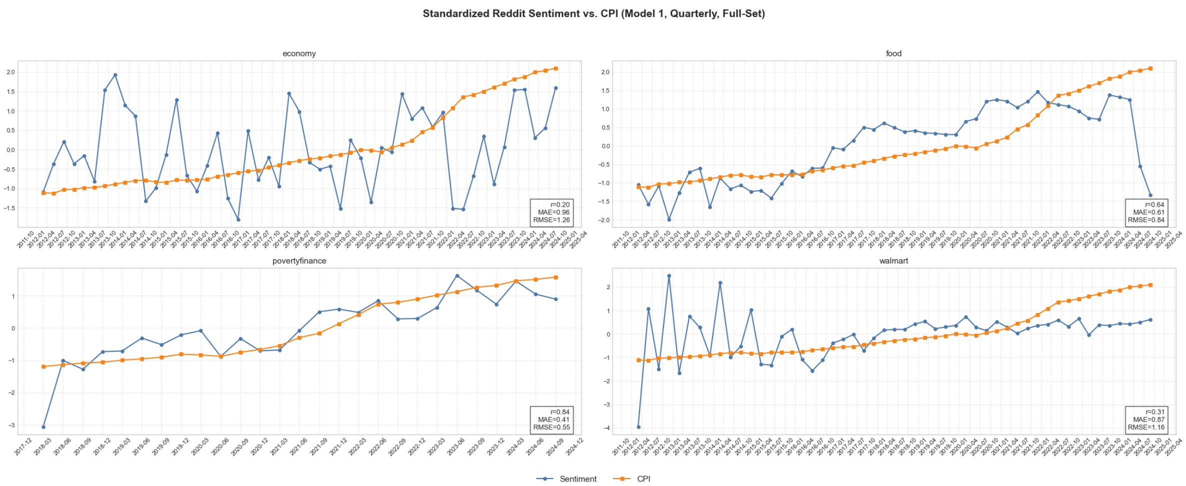

Figure 4: Standardized sentiment (blue) vs. standardized CPI (orange) at quarterly resolution

- **povertyfinance**: $r = 0.84$, MAE $= 0.41$, RMSE $= 0.55$. Strongest positive correlation across all four, even slightly improved from development set values. It should also be mentioned that the subreddit has only existed on Reddit since 2018.
- **walmart**: $r = 0.31$, MAE $= 0.87$, RMSE $= 1.16$. Lower correlation than in the development set, but still positive alignment.

Based on the figures and graphical analysis, `food` and `povertyfinance` perform best. Compared to the others, both curves rise relatively linearly, with food experiencing a slight dip at the end, which does not correspond to the real CPI. This dip shows that these data should be viewed with caution on a monthly aggregate basis and that a good historical trend does not guarantee that this will continue.

4.1.3 Semiannual Aggregation

Figure 5 shows the sentiment–CPI alignment when aggregating snapshots semiannually for each year. This temporal smoothing reduces short-term volatility, particularly in subreddits with high month-to-month variance such as `economy` and `walmart`.

Compared to the quarterly results (see Figure 4), the correlations for `food` ($r = 0.57$) and `Walmart` ($r = 0.53$) remain similar, suggesting that their signal is relatively stable across all time aggregation scales. In contrast, `povertyfinance` increases from $r = 0.84$ to $r = 0.90$ with a low RMSE of 0.45, suggesting that semi-annual

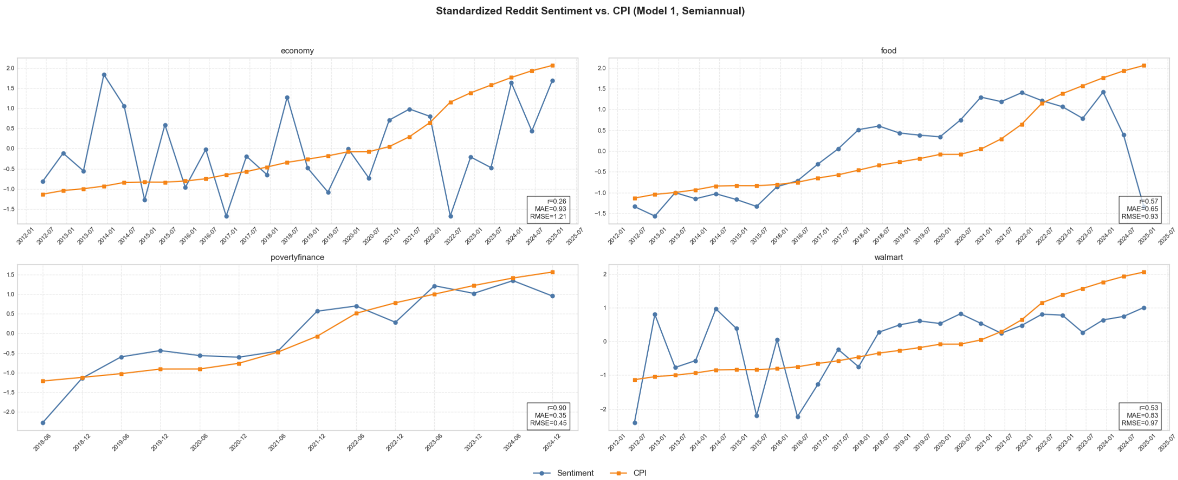

Figure 5: Standardized sentiment (blue) vs. standardized CPI (orange) at quarterly resolution.

aggregation further improves alignment with CPI trends. The subreddit `economy` shows a weaker correlation ($r = 0.26$) compared to its quarterly value ($r = 0.20$), suggesting that smoothing reduces noise but does not significantly improve signal quality in this case.

Overall, the semiannual view confirms expected patterns: subreddits with inherently smoother sentiment trajectories (e.g., `povertyfinance`, `food`) benefit the most from temporal aggregation, while those with high short-term volatility (`economy`, `walmart`) show only marginal gains in correlation.

4.1.4 Annual Aggregation

Figure 6 shows the correlation when using the annual averages of all available monthly snapshots. Compared to the semi-annual aggregation (Figure 5), the correlations for `food` ($r = 0.64$) and `walmart` ($r = 0.64$) remain stable, while the correlation for `povertyfinance` improves slightly to $r = 0.94$ and that for `economy` rises from $r = 0.26$ to $r = 0.36$. Overall, annual aggregation leads to slightly higher correlations for some subreddits, but at the expense of temporal resolution, which can obscure short-term predictive signals. However, the dip in `food` inspected in the quarter cannot be fully eliminated by further aggregations, but it can be somewhat mitigated.

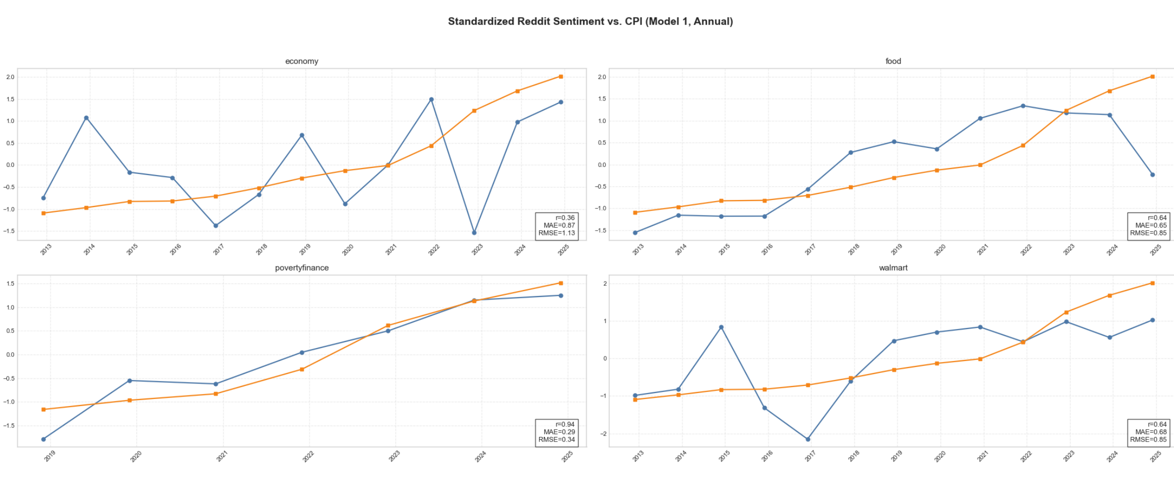

Figure 6: Standardized sentiment (blue) vs. standardized CPI (orange) at quarterly resolution

4.1.5 Summary of VADER Results

Across all time aggregations, the VADER-based sentiment index shows that the selection of subreddits is a key factor in the strength of correlation with official CPI values. It has also been shown that temporal aggregation influences the results: quarterly and semi-annual observations capture more short-term fluctuations but are susceptible to noise, while annual averaging smooths volatility and can improve correlations for stable, domain-oriented subreddits (e.g., `povertyfinance`). These results justify focusing on the best-performing subreddits for subsequent model development, thereby reducing computational effort while retaining the most meaningful signals for inflation. Based on correlation strength and stability, the two best-performing subreddits`food` and `povertyfinance` were selected for further modeling. This choice reflects the trade-off between computational effort and the project goal of systematically identifying the most suitable configuration for robust, interpretable extraction of inflation signals.

4.2 BERT Model (Score Supervision)

4.2.1 Quarterly aggregation

Figure 7 shows the standardized sentiment index (blue) and the standardized CPI (orange) for both subreddits in quarterly resolution. For `povertyfinance`, BERT achieves an exceptionally high correlation ($r = 0.95$), outperforming the VADER baseline ($r = 0.84$) while achieving significantly lower MAE and RMSE values. This suggests that the weakly supervised BERT classifier is able to extract a sentiment inflation signal that almost perfectly matches the CPI fluctuations in this subreddit. For `food`, the correlation is also strong ($r = 0.82$) and improves significantly over the VADER baseline ($r = 0.64$), with the RMSE dropping from 0.84 to 0.61.

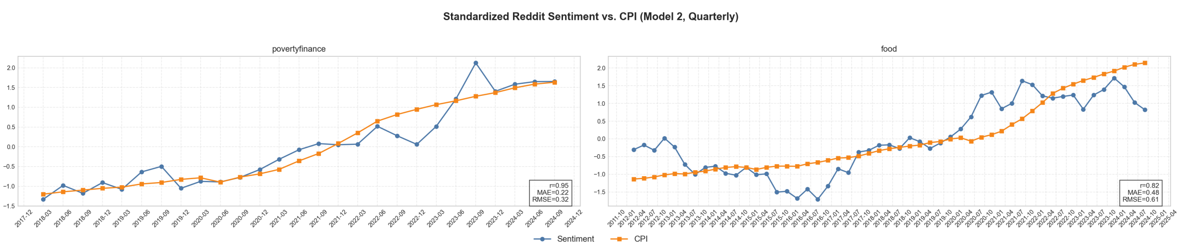

Figure 7: Model 2 (BERT Score Supervision) – Quarterly results for `povertyfinance` and `food`.

Across all three time resolutions, BERT outperforms the VADER baseline in both subreddits, often by a significant margin in terms of r, MAE, and RMSE. This shows that incorporating Reddit-specific language patterns through weak supervision enables more accurate sentiment extraction that aligns with economic signals. The improvement is particularly notable for `povertyfinance`, which shows near-perfect quarterly agreement with the CPI. The graph shows that stagnation and upturns are also recognized by the model (Weak growth in the first half vs. strong growth in the second half).

4.2.2 Semiannual Aggregation

Figure 8 shows the semi-annual aggregation results for the subreddits `povertyfinance` and `food`. The correlation remains very strong for `povertyfinance` ($r = 0.97$, MAE $= 0.15$, RMSE $= 0.24$) and high for `food` ($r = 0.82$, MAE $= 0.46$, RMSE $= 0.60$), with only a slight decrease compared to the quarterly resolution. This suggests that both subreddits exhibit a stable relationship between sentiment and inflation, even when short-term fluctuations are smoothed out. Compared to the VADER baseline (Figure 5), BERT achieves higher correlations and significantly lower error metrics in both cases. The improvement is particularly noticeable in `povertyfinance`, where r increases from 0.90 to 0.97 and MAE decreases from 0.35 to 0.15, suggesting that the supervised approach captures sentiment patterns that better align with CPI movements.

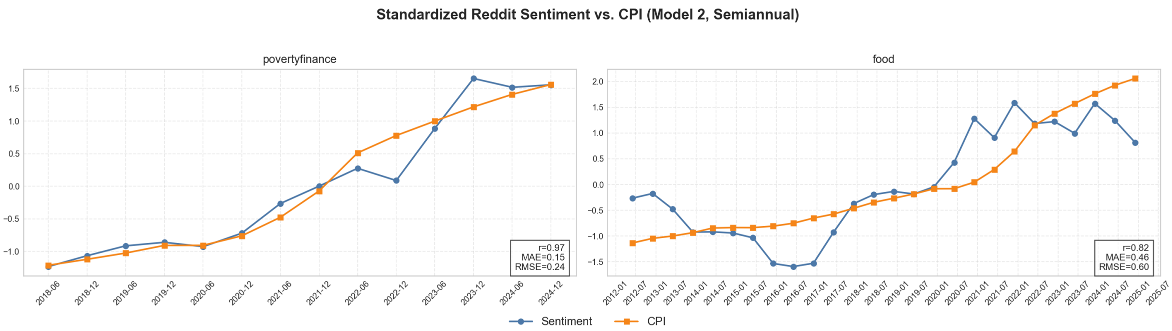

Figure 8: BERT model (score supervision) – semiannual aggregation

4.2.3 Annual Aggregation

Figure 9 shows the annually aggregated sentiment values from the BERT score supervision model. For `povertyfinance`, the annual resolution results in an exceptionally high correlation ($r = 0.98$) with minimal errors (MAE = 0.13, RMSE = 0.20), suggesting that the predictive correlation between sentiment and inflation remains nearly perfect even after maximum temporal smoothing. The subreddit `food` also shows a strong positive correlation ($r = 0.84$) with moderate error values (MAE = 0.46, RMSE = 0.57).

Compared to the annual results from VADER (see Figure 6), BERT consistently delivers higher correlation coefficients and significantly lower MAE/RMSE values, underscoring its superior ability to capture long-term sentiment-inflation trends. The improvement is particularly pronounced for `povertyfinance`, where VADER's value of $r = 0.94$ increases to $r = 0.98$ with BERT, and the MAE decreases by more than 50%. This robustness of performance across all aggregation levels confirms BERT's suitability for extracting stable macroeconomic sentiment signals.

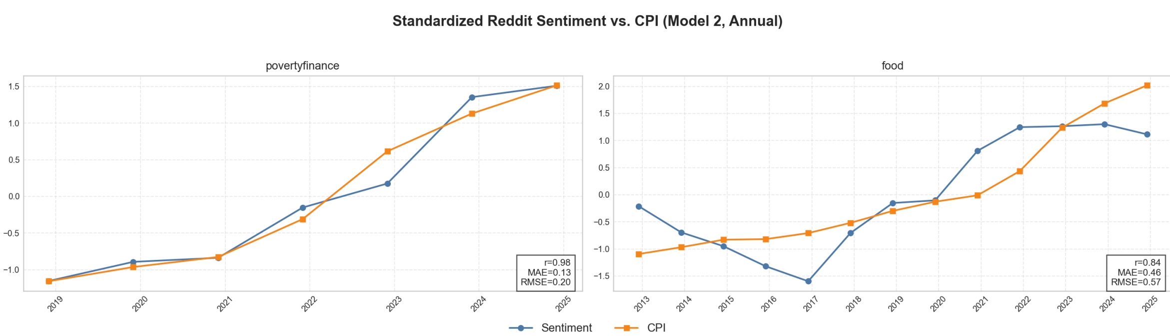

Figure 9: Standardized BERT sentiment vs. CPI (Annual Means) for `povertyfinance` and `food`.

4.2.4 Summary of BERT (Score Supervision) Results

Across all temporal aggregation levels, the BERT Score Supervision model consistently outperforms the VADER baseline in both correlation strength and error metrics for the selected subreddits `povertyfinance` and `food`. The `povertyfinance` subreddit demonstrates exceptional robustness, with correlations exceeding $r = 0.95$ at all resolutions and minimal errors, confirming its strong alignment with CPI trends. The `food` subreddit also achieves high correlations ($r \geq 0.82$) with moderate error levels, indicating stable predictive capacity.

Performance gains over VADER are most pronounced in the annual aggregation, where smoothing generally reduces correlation for all models. Here, BERT maintains or improves r compared to quarterly values while significantly lowering MAE and RMSE, particularly for `povertyfinance` (MAE reduction of more than 50% relative to VADER). These findings confirm that focusing subsequent model development on these two subreddits maximizes predictive signal quality while minimizing computational cost. The results also highlight the systematic benefit of supervised fine-tuning over lexicon-based sentiment scoring for capturing macroeconomic sentiment trends.

4.3 BERT Model (CPI Regression)

4.3.1 Quarterly Aggregation

Figure 10 shows the quarterly CPI regression results for `povertyfinance` and `food` using model 3. Compared to the score-monitored BERT model (and the VADER baseline, the performance is mixed: For `povertyfinance`, the correlation drops sharply from $r = 0.95$ (Model 2) to $r = 0.29$ (Model 3), with MAE and RMSE increasing significantly, indicating weaker temporal alignment between the predicted sentiment and the CPI. For `food`, the correlation drops sharply ($r = 0.68$ compared to $r = 0.82$ in Model 2), with similar error sizes, albeit still slightly worse.

This pattern suggests that direct CPI regression may be more sensitive to data constraints and less robust for smaller, noisy subreddits. The better results of model 2 underscore the advantage of a two-step approach, in which sentiment is first learned and then externally related to the CPI, rather than predicting the CPI directly from the raw text. It is also interesting to note that, similar to the Vader model, there is a sharp drop at the end. Something in the data leads the model to conclude that food inflation is falling sharply. It is also noticeable that while `povertyfinance` performed better than `food` for Model 2, the opposite is true for Model 3. Therefore, it cannot be said per se that `povertyfinance` is better.

Compared to the VADER baseline, model 3 yields also to mixed results: While it slightly improves the correlation for `food` ($r = 0.68$ vs. $r = 0.64$), it performs worse for `povertyfinance` ($r = 0.29$ vs. $r = 0.84$).

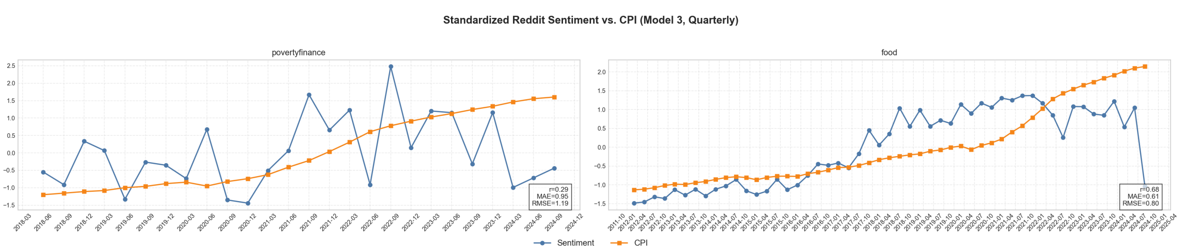

Figure 10: Model 3 quarterly CPI regression results for `povertyfinance` and `food`.

4.3.2 Semiannual Aggregation

Figure 11 shows the semi-annual aggregation results. For `povertyfinance`, model 3 shows a significant decrease in correlation ($r = 0.35$) compared to model 2 ($r = 0.97$) and also lags behind VADER ($r = 0.90$), with the error metrics increasing accordingly. This suggests that the regression-based fine-tuning approach captures a certain trend structure, but loses much of the strong agreement observed in the supervised score variant. For `food`, model 3 performs slightly better than VADER ($r = 0.61$ vs. $r = 0.57$), but lags behind model 2 ($r = 0.82$), suggesting that CPI regression training struggles to maintain high correlation once the data is aggregated semi-annually.

4.3.3 Annual Aggregation

The annual aggregation results reflect the patterns observed in the quarterly and semi-annual analyses. `povertyfinance` continues to show a remarkably strong correlation between sentiment and the CPI, while `food` shows a moderate but stable correlation.

4.4 Summary of all Results

The results show some significant differences in model performance. `povertyfinance` consistently shows higher correlations and lower error metrics compared to `food`, indicating that sentiment trends in `povertyfinance`

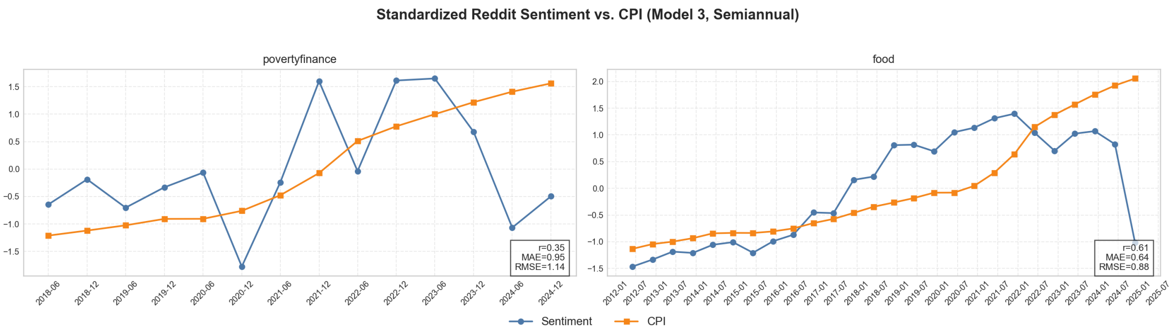

Figure 11: Semiannual aggregation results for `povertyfinance` and `food` under Model 3 (BERT CPI Regression).

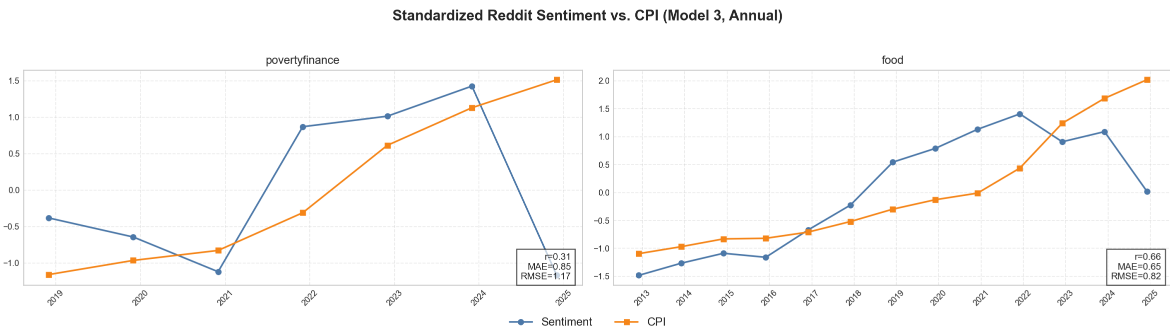

Figure 12: Model 3 (BERT CPI Regression) — Annual aggregation for `povertyfinance` and `food`.

are more strongly and stably linked to CPI changes.

- **Model 1** lies in the middle: it reliably outperforms Model 3 but does not match Model 2. For `povertyfinance`, it achieves strong correlations ($r = 0.84$–0.94) with moderate errors (e.g., annual MAE $= 0.29$, RMSE $= 0.34$). For `food`, the signal is weaker and more variable ($r = 0.57$–0.64; MAE/RMSE around 0.61–0.93), suggesting that direct CPI monitoring is beneficial when the discourse is strongly macroeconomic-budget-oriented, but less so when topics are more heterogeneous.
- **Model 2** provides the best overall match with the CPI. For `povertyfinance`, the correlations are above 0.95 at all aggregation levels (up to $r = 0.98$ annually) with very low errors (MAE ≤ 0.22, RMSE ≤ 0.32). `food` also benefits from this monitoring, achieving $r = 0.82$–0.84 and significantly lower MAE/RMSE values than the other models.
- **Model 3** is the weakest overall. For `povertyfinance`, it shows only weak associations ($r \approx 0.29$–0.35) with large errors (MAE ≈ 0.95, RMSE ≈ 1.14). For `food`, it achieves moderate correlations ($r \approx 0.61$–0.68), but still lags behind the Transformer models in terms of error rate.

Among all experiments, the combination of `povertyfinance` and model 2 (BERT score monitoring) shows the highest potential. Not only does it achieve the strongest correlation with the consumer price index, but it also closely follows actual inflation patterns in the charts and captures both subtle and pronounced upturns. In contrast, models 1 and 3 lead to a sharp decline in the `food` sentiment index at the end of 2024, which is not reflected in the official data. Model 2 successfully avoids this error. One possible reason for this strong performance could be that posts in `povertyfinance` are often directly related to financial difficulties and price sensitivity, making them very indicative of inflation sentiment. Combined with supervised training on Reddit's own point dynamics, Model 2 is able to extract more accurate economic signals from contextually relevant discourse.

Model	Aggregation	povertyfinance			food		
		r	MAE	RMSE	r	MAE	RMSE
Model 1	Quarterly	0.84	0.41	0.55	0.64	0.61	0.84
	Semiannual	0.90	0.35	0.45	0.57	0.65	0.93
	Annual	0.94	0.29	0.34	0.64	0.65	0.85
	Avg.	**0.89**	**0.35**	**0.45**	**0.62**	**0.64**	**0.873**
Model 2	Quarterly	0.95	0.22	0.32	0.82	0.48	0.61
	Semiannual	0.97	0.15	0.24	0.82	0.46	0.60
	Annual	0.98	0.13	0.20	0.84	0.46	0.57
	Avg.	**0.97**	**0.17**	**0.25**	**0.83**	**0.47**	**0.59**
Model 3	Quarterly	0.29	0.95	1.19	0.68	0.61	0.80
	Semiannual	0.35	0.95	1.14	0.61	0.64	0.88
	Annual	0.31	0.85	1.17	0.66	0.65	0.82
	Avg.	**0.32**	**0.92**	**1.17**	**0.65**	**0.63**	**0.83**
Overall Avg.	–	**0.73**	**0.45**	**0.62**	**0.70**	**0.58**	**0.77**

Table 1: Performance metrics across all models and time aggregations for `povertyfinance` and `food`.

5 Live Dashboard

A live dashboard using Streamlit has been implemented to provide an easily accessible and continuously updated overview of inflation trends on Reddit. The current version of the dashboard includes several features: Users can visualize (normalized) sentiment trends over time, aggregated on a daily, weekly, or monthly basis. For transparency reasons, the amount of data is also visualized based on the selected aggregation. In addition, the latest and most positively rated Reddit posts are displayed in tabular form. Using an update button, users can trigger a data update that retrieves the latest Reddit posts, reprocesses them with the trained BERT sentiment model, and updates the CPI figures. The two best-performing subreddits (`povertyfinance` and `food`) are available as options for the subreddits. In addition, the US inflation index is displayed in the same graph. If no official CPI value is available for the current month, the last available value is carried forward as a constant, as an approximation. The underlying model is the most powerful variant from previous experiments (Model 2), with exactly the same keyword filters and preprocessing steps as in the offline version.

To ensure data completeness, posts from the last three days are updated each time the dashboard is refreshed. This approach reduces the risk of relevant discussions being overlooked due to Reddit's API limitations, as only the last 1000 posts per subreddit are returned. All Reddit timestamps and daily aggregations are based on UTC time. Therefore, the start and end of a day may differ from the user's local time zone. It is important to note that the daily or weekly sentiment values provided by the dashboard have not been explicitly evaluated using real inflation data. The validity of the daily and weekly aggregation therefore remains unexamined for the time being, although the monthly aggregation is close to the real data. Regular updates to the dashboard are also necessary in order to enrich the data for the period that can be evaluated at a later stage. The Reddit API does not allow historical review, which is why the dashboard data cannot currently be evaluated due to a lack of history. To illustrate the dashboard layout and functionalities, two representative screenshots are included in the appendix (see Figures 13 and 14). The readme file in the linked GitHub repository describes how to run the dashboard locally (6).

6 Discussion & Conclusion

Strengths

The results show that it is possible to create a sentiment-based inflation index from Reddit data that strongly correlates with US consumer price index figures. Model 2 achieved a correlation of $r = 0.95$ for monthly predictions (restricted to every third month), with an MAE of 0.22 and an RMSE of 0.32. This accuracy underscores the potential of NLP-based methods for extracting economically relevant signals from user-generated text. When the time aggregation window is reduced from one month to two months (half-yearly average of the quarterly months), performance improves further, reaching $r = 0.97$ and MAE $= 0.15$, with a corresponding RMSE of 0.24. This suggests that moderate smoothing can reduce noise.

Future Work and Research Directions

While this project focused on investigating the theoretical feasibility of reconstructing an inflation index from Reddit data, there are still a number of unexplored variables and methodological variations that can be explored in future research. Given that the chosen approach has already achieved strong performance, it was deemed unnecessary to invest significantly more computing and storage resources for marginal potential gains within the scope of this project. Nevertheless, several promising variations remain open. Future work could explore the effects of adjusting the minimum value for post evaluation (both upward and downward) and changing the minimum number of comments. Another extension concerns the refinement or expansion of the keyword sets used for filtering. The inclusion of additional subreddits would help to expand the thematic coverage. It is also possible to apply this to other countries with appropriately selected subreddits.

On the modeling side, experiments with hyperparameters such as learning rate or batch size could provide information about trade-offs between efficiency and accuracy. Comparing Reddit-based indices with sentiment indicators from other platforms such as X, Instagram, or Facebook would also help to assess cross-platform consistency. A deeper analysis of the texts, combined with inflation figures, could also provide new insights that could improve the model. For example, by analyzing the texts in months with high and low inflation. In addition, the dashboard allows for the continuous enrichment of live data over time. This makes it possible to retrospectively evaluate the accuracy of daily and weekly index values and to test which model configurations perform best for different aggregation levels.

Limitations

One limitation that must be taken into account is the characteristic of the CPI series, which exhibits a linear upward trend. This characteristic favors models that provide similarly smooth and monotonic predictions, which may inflate the correlation values without necessarily capturing the actual underlying economic dynamics. However, as shown in Figure 8, the model also follows certain local deviations, particularly in `povertyfinance` in earlier years, suggesting that it does not exclusively follow the global trend. It should also be noted that the official CPI figures are only an approximation of the actual currency devaluation, as they are based on korba-based estimates. Additionally, the live dashboard has not yet undergone formal evaluation for its daily or weekly sentiment values. The real-time index should therefore be interpreted as an experimental approximation until a thorough validation over longer time periods has been conducted.

Conclusion

This study demonstrates that it is feasible to construct a Reddit based sentiment index that closely mirrors official U.S. inflation trends. Among the models tested, the score-monitored BERT model achieved the highest correlations with the CPI, especially for the subreddit `povertyfinance`, where the correlations were above $r = 0.95$ with minimal MAE and RMSE. Visualizations further support the results, illustrating that the best model successfully tracks both subtle and pronounced inflation changes. The results confirm that thematically filtered online discussions can serve as a robust substitute for macroeconomic indicators. They also show that the selection of subreddits and moderate temporal aggregation are crucial for maximizing predictive accuracy. These findings provide a strong empirical basis for further exploration of social media–based economic indices and their application in real-time economic monitoring.

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Author's Note on AI Assistance

Parts of the text in this report have been reworded using ChatGPT (OpenAI) to improve readability and style. In addition, DeepL was used for some translations between German and English.

Code Availability

All scripts and data preprocessing pipelines used in this study are publicly available at: `https://github.com/christianleis040/reddit_inflation_index`.

Appendix

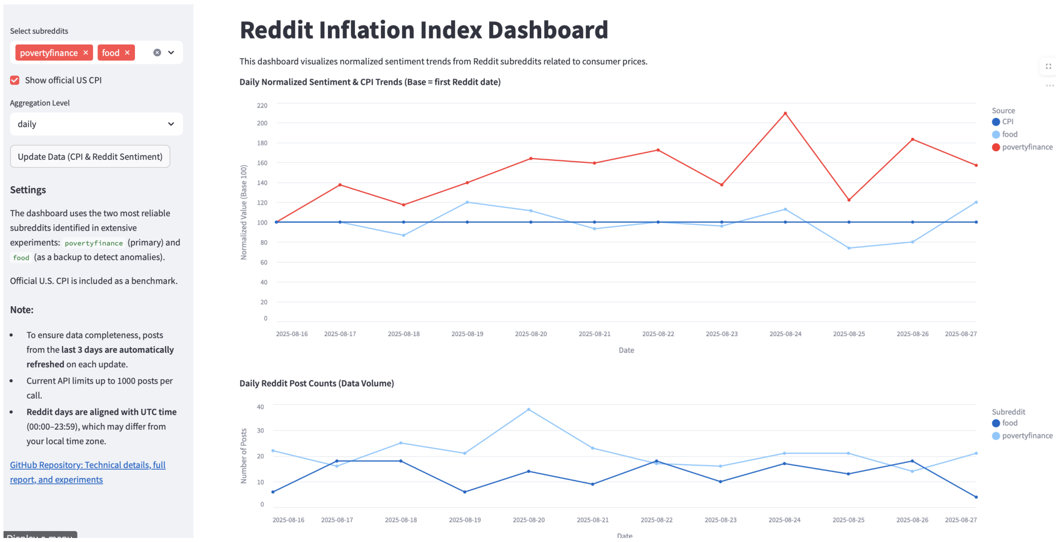

Figure 13: Main view of the Reddit Inflation Index dashboard showing normalized (27.08.2016).

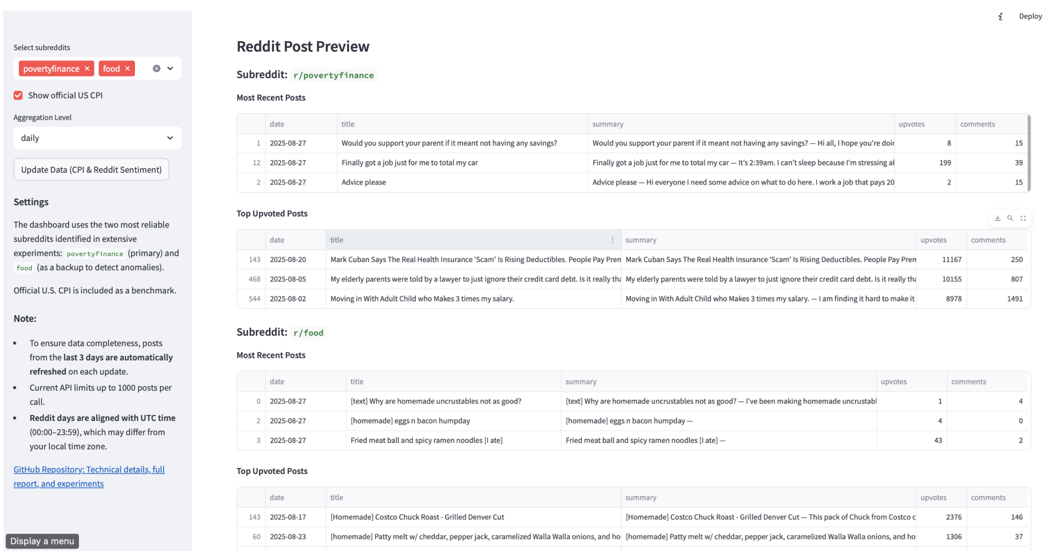

Figure 14: Table view with recent and top-upvoted Reddit posts per subreddit (27.08.2016).